Mod extension PCB/prototype board

Center negative
18V–36V
J3
RLED 10k
D6 LED
GND
D5 1N5819
C17 100n film
C18 47u electrolytic
R25 100R
C19 100n film
C20 100u electrolytic
R28 100R
D7 39V/1.3W
C21 100n film
C30 100u electrolytic
GND
R30 10k
Q2 2N3904
VOLTS2 A1M
R31 1k
R33 220R
3V–36V
C23 47u electrolytic
R45 47k
R46 47k
C37 47u electrolytic
R26 10k
Q1 2N3904
VOLTS1 A1M
R27 1k
R29 220R
C22 47u electrolytic
SW1 CLIP1
QC1 BS170
QC2 BS170
SW2 CLIP2
QC3 BS170
QC4 BS170
R34 100R
C25 100n film
C24 10p ceramic
R35 470k
Q3 2N3904
R37 10k
C26 4u7 electrolytic
R38 2k2
TAME A100k
C27 3n3 film
C28 100n film
R39 470k
Q4 2N3904
R41 10k
C28 4u7 electrolytic
R42 100R
C29 10n ceramic
R43 10k

Check your BS170s pinouts first!
They may vary a lot. Don't just google it,
use for example a transistor tester instead.

Main/stock Muff PCB

18V
C14 100u electrolytic
GND
R4 22k
C2 560p
R3 390k
C3 100n
J1 Input jack
IN
T
SN
S
R1 33k
C1 100n
R23 1M
C15 10p ceramic
ADDED
2n2369a hFE 40-60
T1
R2 82k
R5 820R
C33 10u
ADDED
A100k Gain
C4 100n
R6 1k
R7 8k2
R8 82k
R9 390k
C6 560p
C7 100n
R10 12k
R11 150R
C34 10u
ADDED
R12 8k2
R13 82k
R14 390k
C9 560p
C8 47n
R15 22k
R16 820R
C35 10u
ADDED
T2
T3
C5 47n
C10 3n9
R18 39k
R17 39k
C11 10n
Tone B100k
C12 100n
R20 390k
R21 12k
R19 100k
R22 2k7
C36 10u
ADDED
T4
C13 100n
R24 100R
C16 100p ceramic
ADDED
Volume A100k
J2 Output jack
T
S

Cut the trace using a scalpel
SPLIT
Don't solder D1 & D2 (or cut them off).
REMOVED
D1 1n4148
D2 1n4148
D3 1n4148
D4 1n4148

Optional:
After cutting the traces you can tap the spot on the PCB where you made a cut with a nail polish to avoid metal corrosion.

Or you can avoid cutting the traces by not soldering top ends of R10 and R15, instead wiring them to the extension board.

Do not solder the pots onto PCB, use wires. Use GAIN/SUSTAIN pot pin 3 PCB hole to add a jumper wire to pin 2 hole of the TONE pot. Thus sending the signal from T1 straight to T4.

Do not solder the pots onto PCB, use wires. Use PCB TONE pot hole 2 to add a jumper to receive the signal from T1. While TONE pot pin 2 is connected to the VOLUME pot pin 3 (bypassing PCB).

Do not solder VOLUME pot onto PCB at all. Instead Use PCB VOLUME pin holes to add the RC filter. Solder R24 across pin 3 and 2 holes, and solder C16 across pin 2 and 1 PCB holes. And solder a jumper wire to PCB OUT (which is shorted with VOLUME pot PCB pin 2 hole). And that jumper wire goes to GAIN pot pin 3 (bypassing PCB). Thus your T1 goes to T4 and then goes into the GAIN pot before T2.

You don't need to cut anything here, just make sure VOLUME pot pin 2 goes to extension mod PCB input, and extension mod PCB output goes to the output jack.

N.B.
Musikding.de parts list has an error
R1 is supposed to be 33k, not 3k3.

This mod:

1. Removes clipping diodes completely, letting the transistors clip instead, making classic Muff circuit loud and wild, more dynamic, like a real fuzz pedal.

 At the same time it provides optional/switchable BS170 MOSFET pairs as clipping diodes replacement. The forward voltage is much higher, ≈2.60V (for each direction) vs. ≈0.7V for regular 1n4841. So even though there's no make-up gain stage you would still have plenty of output volume on tap because of that. The knee is also significantly different from regular diodes, you get a non-linear, interesting distortion character with a rounder/softer clipping transitions.

2. Re-routes T4 gain stage to go after T1, giving the clipping stages more pre-gain as they require much more to really clip, since there are no diodes, only transistors clipping when they can amplify no more. And you don't need the final VOLUME boost at all, since the clipping stages without the clipping diodes go LOUD!

3. VOLUME knob goes right after the TONE knob after the clipping stages, and then the signal goes to a unity-gain buffer.

4. Adds buffered rat-style tone control (TAME knob, low-pass filter) on top of existing Muff TONE control that allows to tame harshness in the highs with extreme Muff's TONE settings, adding more usable tonal options (e.g. very tight on the low end but not painfully bright/too splattery).

5. Adds 2x VOLTS knobs that is controlling buffered voltage regulator for each clipping stage individually in the middle of the circuit, ranging from very low, when the transistor is close to around ≈18V/≈36V (≈max supply voltage), allowing you to have kind of a splattery gated fuzz, which also reduces highs and lows content.

6. Accepts from 18V to 36V power supply, higher voltages make pre-boost gain stages cleaner and more high-headroom gain while the clipping stages can be set to whatever voltage you prefer individually (see VOLTS knobs, ≈3V–≈36V). I find 18V to be a sweet spot source voltage.

7. Improves DC filtering and adds extra RF filters, making the circuit less noisy, more stable.

8. Make the pedal output buffered, so your VOLUME knob does not affect output impedance of the pedal, it's always low.

Pots wiring (instead of cutting traces on the PCB):

GAIN/SUSTAIN pot pins 1 and 2 are soldered onto PCB, as designed by the PCB. Pin 3 though is soldered to the PCB TONE pot PCB hole 2 (thus sending the signal from T1 straight to T4).

TONE pot pins 1 and 3 are soldered onto the PCB as designed by the PCB. Pin 2 of the pot though is connected to pin 3 of the VOLUME pot (bypassing the PCB). So that the VOLUME pot is going right after the TONE pot.

None of the VOLUME pot pins are connected to the main PCB. Pin 3 is connected to TONE pot pin 2, pin 1 is soldered to the pot back/case (which is supposed to be grounded by touching the metal of the shielded pedal case). And pin 2 is connected to the extension mod PCB first buffer input (R34).

In the schematic stock parts for Muff circuit are the TRIANGLE one but you can use whatever Muff variant you prefer.
Main Muff schematic is based on Musikding.de Muff DIY kit layout.
https://www.musikding.de/Der-Muff-Triangle-Distortion-kit

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Title: Wenzel's Real Fuzz mod for classic Muff circuit

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